

SUPPLY CHAIN ANALYTICS PROJECTS

★ Supply Chain Analytics Compendium, Six Analytical Cases

2026

Regression, forecasting, ABC classification, push-pull economics, sourcing risk for semiconductor industry

- Worked six supply chain problems end to end and documented each as a case: trend and multiple regression forecasting on a 24-month series (R-squared 0.439 to 0.948, out-of-sample MAPE 10.57%), ABC classification across 36 SKUs, push versus pull inventory economics, and sourcing analysis under supplier concentration.
- Reported weak fits and contradicted rules of thumb rather than only favourable results, including a statistically significant trend model rejected as unusable for planning because its standard error reached half the mean order size.

Google Drive:  [Supply_Chain_Analytics_Compendium.pptx](#)

★ Inventory Brain & Demand Planning Platform

2026

Python, forecasting, optimisation, discrete-event simulation, MLOps, dashboard

- End-to-end planning platform combining demand forecasting, multi-echelon inventory optimisation across distribution tiers, discrete-event simulation of replenishment policies under demand and lead-time variability, and a decision dashboard for scenario comparison.

GitHub: github.com/yzarrai/inventory-brain

★ Demand & Stock Fluctuation Forecasting Machine Learning

2026

Python, SARIMA, Prophet, XGBoost, SHAP

- Benchmarked SARIMA and Prophet against XGBoost under walk-forward validation on MAE, RMSE and MAPE, then converted the selected forecast into service-level-driven safety stock and reorder point recommendations, with SHAP attribution to explain which demand drivers moved each prediction.

GitHub: github.com/yzarrai/retail-demand-forecasting-ml

Procurement RAG Assistant

2026

Python, hybrid search (BM25 + Dense RRF), LLM grounding

- Built an enterprise-grade retrieval-augmented generation assistant for procurement documents with clause-level citations, a 5-layer grounding guardrail system to prevent hallucination, and structured commercial risk scanning across contract text.

GitHub: github.com/yzarrai/procure-rag

Stochastic Cold-Chain Distribution Planner

2026

Python, MILP, stochastic optimization

- Developed a two-stage stochastic optimization and decision-support pipeline for cold-chain vaccine distribution, modeling maritime disruption scenarios to recommend robust distribution plans under uncertainty.

GitHub: github.com/yzarrai/stochastic-coldchain-planner

Multi-Agent Supply Chain Agentic Artificial Intelligence

2025

Python, MILP (PuLP), SARIMAX, DuckDB, ChromaDB, FastAPI, Docker

- Built a 4-agent autonomous swarm (Demand, Inventory, Risk, Logistics) using a 2-round multi-agent debate protocol to reach Pareto-optimal trade-offs, combined with MILP stochastic lot-sizing, SARIMAX forecasting, and a Monte Carlo digital twin for disruption stress-testing.
- Integrated a DuckDB/ChromaDB hybrid RAG data layer with a generative control tower dashboard supporting human-in-the-loop approvals and simulated SAP/Oracle ERP webhook dispatch, validated by 90 automated tests.

GitHub: github.com/yzarrai/supply-chain-agentic-ai

Supply Chain Simulation LLM Agent

2025

Python, SimPy, LangChain, Pydantic

- Built a discrete-event simulation (SimPy) of a 4-echelon supply chain with (s, Q) replenishment policy, using Common Random Numbers to cut Monte Carlo variance by 19–37% for statistically policy comparisons.
- Designed a zero-hallucination LangChain agent constrained to strict tool schemas so every quantitative claim is grounded in simulation output, validated with 39+ automated tests.

GitHub: github.com/yzarrai/supply-chain-simulation-agent

Decision Support & MLOps Systems (Banking, Clinic Operations)

2025

Python, SimPy, pm4py, MLflow, discrete-event simulation

- Built three end-to-end decision support and MLOps systems across loan origination and clinic operations, combining discrete-event simulation, process mining, ML forecasting, and production MLOps (MLflow registry gates, drift monitoring, closed-loop retraining).

GitHub: github.com/yzarrai/loanflow-dss/

github.com/yzarrai/loanflow-mlops / github.com/yzarrai/clinic-flow-dss

CERTIFICATIONS

- Supply Chain Analytics Specialization (6 courses): Analytics Essentials, Business Intelligence & Competitive Analysis, Demand Analytics, Inventory Analytics, Supply Chain Analytics, Sourcing Analytics
- Supply Chain Management Specialization, Rutgers Business School (5 courses): Logistics, Operations, Planning, Sourcing, Management Strategy. Capstone: Medical Technology Corporation
- Process Mining: Data Science in Action, Eindhoven University of Technology
- Health Informatics Specialization, Johns Hopkins University
- Google IT Automation with Python Professional Certificate, Google
- AI and cloud: Google Artificial Intelligence Professional Certificate; LangChain for LLM Application Development and Agentic AI, DeepLearning.AI; Fundamentals of Cloud & Generative AI, Amazon Web Services
- Full Stack Product Management, RevolU

SKILLS

Supply Chain Analytics: Demand planning and forecasting, inventory optimisation, safety stock and reorder point modelling, sourcing and supplier risk analytics, network and facility location optimisation, supply chain KPIs

Supply Chain Operations: Procurement, inventory management, logistics coordination, regulated goods handling, Lean Six Sigma (DMAIC), business process management, BPMN

Data & Analytics: Google Sheets, Python, SQL, Power BI, Tableau, Streamlit, statistical modelling, time series forecasting, machine learning, discrete-event simulation, process mining (Celonis, pm4py)

Optimisation & Automation: Linear and integer programming, p-median facility location, workflow automation, Python automation, MLOps (MLflow, FastAPI, drift monitoring), CI/CD with GitHub Actions, n8n, Zapier, Make.com

Enterprise Systems: Working knowledge of SAP, Oracle, Odoo and Salesforce (CRM) from a solutions and implementation perspective

Project & Product Management: Jira, Notion

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